GARMIN INTERNATIONAL

Garmin Developer API Programs
Women's Health API

Version 2.0

CONFIDENTIAL

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1. Revision History

Version	Date	Revisions
1.0	12/01/2020	Initial revision
1.0.3	10/16/2023	Removing reference for the user access token form PING/PUSH notifications examples in preparation to retire this field.
1.0.5	11/20/2025	API configuration tool description added.
1.0.6	5/1/2026	Production review updated
2.0	10/15/2026	Garmin Connect Developer Program updated

2. Purpose of Women's Health API

The Garmin Connect Women's Health API lets you access information related to menstrual cycle tracking and schedule information. Similar to the Health API, these metrics are also applicable to the wellness population but additionally may be utilized alongside training plans and workout schedules. After user consent, you can access the data logged by end-users.

3. Endpoint Configuration

Women's Health API is server to server communication only. We deliver event driven notifications to your endpoints configured at https://apis.garmin.com/tools/endpoints. Log in using your client ID and client secret.

Each enabled summary should be configured with a valid HTTPS URL to which Ping or Push notifications for that summary type will be sent. Other protocols and non-standard ports are not supported. Please make sure the enabled URLs do exist and accept HTTPS POST requests.

Enabled: When checked, this summary data will be made available for all users associated with this client ID and the summary type will be sent to the provided URL. When unchecked, data will *not* be made available, notifications will not be sent, and any notifications in queue (including failed) will be dropped.

On Hold: When checked, data will continue to be available, but notifications will be queued and not sent. Notifications will be queued for up to seven days and then dropped. When unchecked, all previously queued notifications will be sent serially. If a summary type is not enabled this setting has no effect.

Tip: On Hold functionality is useful for planned maintenance events or any other instance when it would be useful to temporarily stop the flow of notifications without data loss. Although a missed notification will be re-attempted for as long as possible, using On Hold guarantees seven days of availability as well as resumption of notifications within 2 minutes of disabling the setting. Normal resumption time may be longer due to exponential back-off between failed notification re-attempts.

4. Push Service

The Push Service allows 3rd party apps to receive near-real-time updates of Garmin user data without delay or duplication associated with regularly scheduled update jobs. The Push Service generates HTTPS POSTs that contain the updated data directly within the POST as JSON.

The Push Service has a timeout of thirty seconds. To avoid missed data or improper error responses, it is required to respond to each notification with an HTTP status code of 200 (OK) before performing any processing.

4.1 Push Notification Content

JSON Element	Description
summary type (list key)	The summary type of this list of pings.
userId	A unique user identifier corresponding to the underlying Garmin account of the user.
userAccessToken	The UAT corresponding to the user that generated the new data.
Summary data	The summary data in the same data model as the Women's Health API. See the Summary Endpoints section for details and examples of each summary data model.

Example


```
{
   "mct": [
      {
         "userId": "4aacafe82427c251df9c9592d0c06768",
         "summaryId": "x2faf44e-16cfcc8d020",
        "periodStartDate": "2019-09-03",
        "dayInCycle": 1,
        "periodLength": 1,
        "currentPhase": 1,
        "lengthOfCurrentPhase": 1,
        "daysUntilNextPhase": 1,
        "predictedCycleLength": 16,
        "isPredictedCycle": false,
        "cycleLength": 16,
        "lastUpdatedTimeInSeconds": 1567609114
      }
   ]
}
```

5. Women's Health API Integration Tips

This section describes functionality that is important to understand when integrating with the Garmin Connect Women's Health API and tools to help accelerate and verify that integration.

5.1 Time Values in the Women's Health API

All timestamps in the Women's Health API are UTC in seconds, also known as Unix Time. However, summary data records may also contain a time offset value. This value represents the difference between the standardized UTC timestamp.

6. Summary Endpoints

This section provides details of the data available for each summary type. Summary data records are the core method of data transfer in the Women's Health API, with each summary corresponding to a different ping notification type.

6.1 Menstrual Cycle Tracking (MCT) Summaries

The Menstrual Cycle Tracking feature (https://connect.garmin.com/features/menstrual-cycle-tracking/) available on some Garmin devices allows users to track information about their cycle schedule and log symptoms. The MCT Summary only returns information related to cycle schedule (see response parameters) and does not make the information about symptoms available.

In addition to providing consent to share their Garmin Connect data with your Women's Health API program, users must also indicate their permission to share their MCT schedule information.

Request

Each MCT summary may contain the following parameters:

Property	Type	Description
summaryId	string	Unique identifier for the summary
periodStartDate	string	The calendar date representing period start date. The date format is 'yyyy-mm-dd'
dayInCycle	integer	Represents ^{nth} day in cycle
periodLength	integer	Number of days indicating how long a period usually last

currentPhase	integer	Indicates the phase in this cycle like menstruation, fertile, etc.,
currentPhaseType	String	Indicates the phase in this cycle.
lengthOfCurrentPhase	integer	Represents the length of current phase in days
daysUntilNextPhase	integer	Number of days remaining to reach the next predicted phase
predictedCycleLength	integer	Number of days predicted to be the current cycle length
isPredictedCycle	boolean	A boolean to show if this summary is a predicted cycle or not
cycleLength	integer	A user logged cycle length
lastUpdatedTimeInSeconds	integer	Time in seconds showing when a user logged their symptoms
hasSpecifiedCycleLength	boolean	A boolean to show if cycle lenght was provided by the user through Garmin Connect
hasSpecifiedPeriodLength	boolean	A boolean to show if period length value was provided by the user through Garmin Connect

If a user is in a pregnant phase, the pregnancySnapshot field will be populated with the following fields:

Property	Type	Description
title	String	Title used for the user's pregnancy
originalDueDate	String	The calendar date representing the user's original due date entered by the user
dueDate	String	The calendar date representing an updated due date. This value is entered by the user in Garmin Connect.
pregnancyCycleStartDate	String	The calendar date representing the start of the user's pregnancy cycle
numOfBabies	String	A string representing the number of babies during the user's pregnancy
weightGoalUserInput	Dictionary	A representation of the user's height in centimeters (heightInCentimeters) and weight in grams (weightInGrams). This is entered by the user when the user is marked as pregnant in Garmin Connect. Please see below example for data structure.
bloodGlucoseList	Dictionary	A representation of the user-logged blood glucose readings. This contains valueInMilligramsPerDeciliter (floating point), logType (String), and reportTimestampInSeconds (integer). Please see below example for data structure.

Example

Example where user is marked as pregnant

[

```
{
    "summaryId": "x2faf44e-175f69ac4fa",
    "periodStartDate": "2020-10-11",
    "dayInCycle": 110,
    "currentPhase": 6,
    "currentPhaseType": "SECOND_TRIMESTER",
    "lengthOfCurrentPhase": 98,
    "daysUntilNextPhase": 80,
    "predictedCycleLength": 280,
    "isPredictedCycle": false,
    "cycleLength": 280,
    "lastUpdatedTimeInSeconds": 1606160139,
    "hasSpecifiedCycleLength": false,
    "hasSpecifiedPeriodLength": false,
    "pregnancySnapshot": {
        "title": "Pregnancy 2021",
        "originalDueDate": "2021-07-17",
        "dueDate": "2021-07-17",
        "pregnancyCycleStartDate": "2020-10-11",
        "numOfBabies": "SINGLE",
        "weightGoalUserInput": {
            "heightInCentimeters": 162,
            "weightInGrams": 58966
        },
        "bloodGlucoseList": [
            {
                "valueInMilligramsPerDeciliter": 97.0,
                "logType": "BEFOREBED",
                "reportTimestampInSeconds": 1607032423
            },
            {
                "valueInMilligramsPerDeciliter": 92.0,
                "logType": "BEFOREMEAL",
                "reportTimestampInSeconds": 1607032403
            },
            {
                "valueInMilligramsPerDeciliter": 89.0,
                "logType": "AFTERMEAL",
                "reportTimestampInSeconds": 1607356249
            },
            {
                "valueInMilligramsPerDeciliter": 93.0,
                "logType": "AFTERMEAL",
                "reportTimestampInSeconds": 1607523916
            },
            {
                "valueInMilligramsPerDeciliter": 86.0,
                "logType": "OTHER",
                "reportTimestampInSeconds": 1607536280
            },
            {
                "valueInMilligramsPerDeciliter": 50.0,
```

```
                    "logType": "BEFOREMEAL",
                    "reportTimestampInSeconds": 1611852498
                }
            ]
        }
    }
]
```


Example where user is in non-pregnant phase


```
[
    {
        "summaryId": "x153a9f3-176e4715000",
        "periodStartDate": "2021-01-04",
        "dayInCycle": 1,
        "periodLength": 5,
        "currentPhase": 1,
        "currentPhaseType": "MENSTRUAL",
        "lengthOfCurrentPhase": 5,
        "daysUntilNextPhase": 5,
        "fertileWindowStart": 11,
        "lengthOfFertileWindow": 7,
        "predictedCycleLength": 28,
        "isPredictedCycle": true,
        "cycleLength": 28,
        "lastUpdatedTimeInSeconds": 1610150400,
        "hasSpecifiedCycleLength": false,
        "hasSpecifiedPeriodLength": false,
        "pregnancySnapshot": {}
    }
]
```

```
[
    {
        "summaryId": "x153a9f3-176e4715000",
        "periodStartDate": "2021-01-04",
        "dayInCycle": 1,
        "periodLength": 5,
        "currentPhase": 1,
        "currentPhaseType": "MENSTRUAL",
        "lengthOfCurrentPhase": 5,
        "daysUntilNextPhase": 5,
        "fertileWindowStart": 11,
        "lengthOfFertileWindow": 7,
        "predictedCycleLength": 28,
        "isPredictedCycle": true,
        "cycleLength": 28,
        "lastUpdatedTimeInSeconds": 1610150400,
        "hasSpecifiedCycleLength": false,
        "hasSpecifiedPeriodLength": false,
        "pregnancySnapshot": {}
    }
]
```


7. Enterprise Level – Historical data

This service provides the ability to request historic summary data for a user. Historic data, in this context, means any data uploaded to Garmin Connect before the user's registration with the 3rd party app.

A historical data request returns an empty HTP 202 response immediately, while the actual process takes place asynchronously in the background. Once request is complete, a notification will be generated and sent as if data for that time period was newly synced. The maximum date range (inclusive) for a single request is 1 month (from today).

* Note: Duplicate historical data requests are rejected with HTTP 409 status (duplicate requests – requests for already requested time period)

Rate limits:

User rate limit – **1 month**

Client ID rate limit – 10,000 days/minute

Resource URL for Women's Health summaries

GET https://apis.garmin.com/wellness-api/rest/backfill/mct

Request parameters

Parameter	Description
summaryStartTimeInSeconds	A UTC timestamp representing the beginning of the time range to search based on the moment the data was recorded by the device. This is a required parameter.

summaryEndTimeInSeconds	A UTC timestamp representing the end of the time range to search based on the moment the data was recorded by the device. This is a required parameter.

Response: HTTP 202
Historical data request works asynchronously, a successful request returns HTTP status code 202 (accepted) with no response body. Please see Appendix B for possible error responses.

Request:
GET https://apis.garmin.com/wellness-api/rest/backfill/mct?summaryStartTimeInSeconds=1452384000&summaryEndTimeInSeconds=1453248000

This request triggers the historical data for summary records which were recorded in the time between UTC timestamps 1452384000 (2016-01-10, 00:00:00 UTC) and 1453248000 (2016-01-20, 00:00:00 UTC).

8. Enterprise Level – Ping/Pull Service

Garmin will send HTTPS POST ping notifications regarding the availability of new summaries and de-registrations to licensee shortly after new data is available. This Ping Service allows app to maintain near-real-time consistency with the Garmin data store without wasted queries on users that haven't synced any new data.

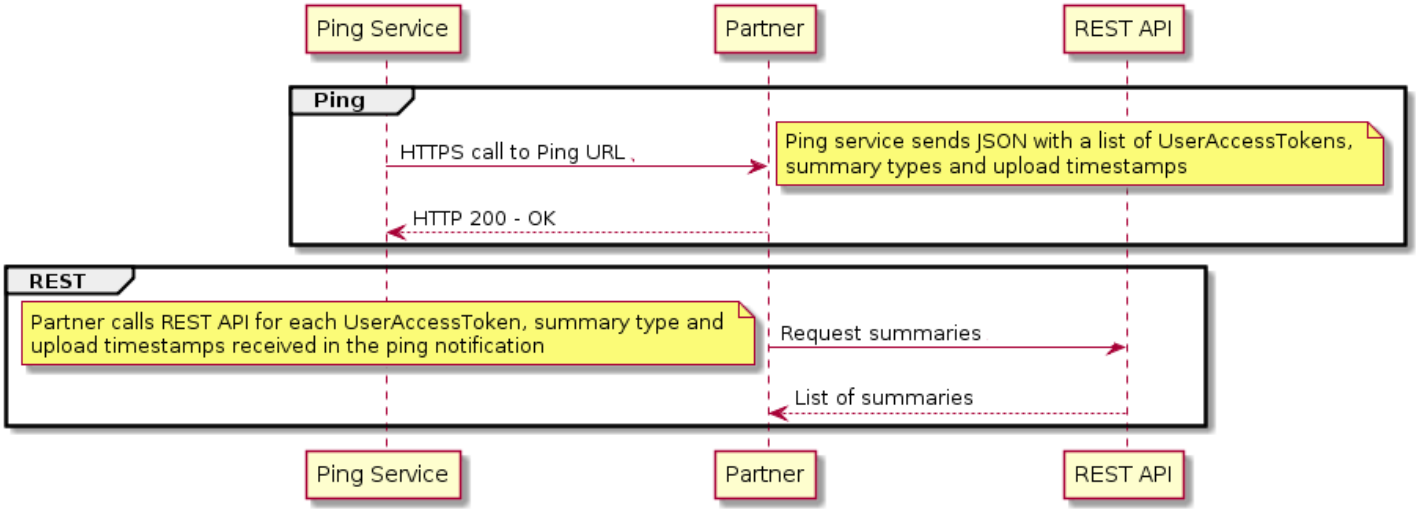
Each notification also contains a callback URL. When this URL is called, data specific to that user and summary type is returned. The licensee may provide separate URLs for each summary type for flexible processing or may choose to send ping notifications for all data types to the same endpoint.

Tip: Please call the Women's Health REST API asynchronously after closing the connection of the ping request. One frequent ping/pull implementation mistake is to hold the incoming ping notification HTTP POST open while performing the corresponding the callbacks to the Women's Health API. This will result in HTTP timeouts and pootential data loss.

Each ping message contains a JSON structure with a list of userIDs for which new data is available, as well as the URL to call to fetch that data. A successful ping-based integration should never need to call the Women's Health API except as prompted by ping notifications.

8.1 Ping Workflow

The following diagram illustrates the general workflow.



The Ping Service has a timeout of thirty seconds. In order to avoid missed data or improper error responses, it is required to respond to each notification with an HTTP status code of 200 (OK) before performing callbacks to the Women's Health API. Holding the ping open while performing callbacks is the most common cause of instability in Women's Health API integrations.

A failed ping notification is defined as any of the following:

- The app's ping endpoint is unreachable
- The endpoint responds with an HTTP status code other than 200
- An error occurs during the request (e.g. the connection breaks)

In the case of a failed ping notification, the Ping Service attempts to re-send the ping on a regular basis. The Ping Service will continue to re-attempt failed pings, successively waiting longer between each attempt, for as long as the failed ping queue depth does not affect the performance of the overall Women's Health API.

Tip: If you know in advance that your notification end points will be unavailable (e.g. server maintenance), you may set your notification to "On Hold" using the Ping Configuration Web Tool (see Web Tools below). Doing so will guarantee quick transmission of pings once the on-hold state is removed and avoid data loss.

8.2 Ping Notification Content

JSON Element	Description
summary type (list key)	The summary type of this list of pings
userId	A unique user identifier corresponding to the underlying Garmin account of the user. This userId is *not* used as a parameter for any call to the Women's Health API. However, it will persist across userAccessTokens should the user re-register to generate a new UAT.
callbackURL	Pre-formed URL to pull the data. Not present for deregistration notifications.

Example

```
{
      "mct": [{
            "userId": "4aacafe82427c251df9c9592d0c06768",
            "callbackURL": "https://apis.garmin.com/wellness-
api/rest/mct?uploadStartTimeInSeconds=1444937651&uploadEndTimeInSecond
s=1444937902"
      }]
}
```

Tip: During your Ping Service integration development, it may be cumbersome for your endpoints to be publicly available to receive real notifications from the API. Simulating ping requests within the local network by using tools like cURL is a useful way to solve this problem. You can generate a temporary pull token via API Pull Token Tool.

Here is an example for simulating a ping request for epoch summaries for a service running on localhost, port 8080:

```
curl -v -X POST "http://localhost:8080/garmin/ping" \
-H "Content-Type: application/json;charset=UTF-8" \
-H "user-agent: Garmin Health API" \
-H "accept: text/plain, application/json, application/*+json, */*" \
-H "accept-encoding: gzip, x-gzip, deflate" \
-H "garmin-client-id: <CLIENT_ID>" \
-H "x-forwarded-for: 204.77.163.244" \
-H "x-forwarded-proto: https" \
-d '{
  "mct": [
    {
      "userId": "<USER_ID>",
      "callbackURL": "https://apis.garmin.com/wellness-
api/rest/mct?uploadStartTimeInSeconds=<START_TIME>&uploadEndTimeInSeco
nds=<END_TIME>&token=<TOKEN>"
    }
  ]
}'
```

Appendix A – Error Responses

Usually the service responds to all requests with HTTP status code 200 (OK). In case of an error, one of the following HTTP status codes may be sent. When any of these HTTP status codes are present, the response body will contain a JSON object with an error message to assist in isolating the exact reason for the error in the following form:

{"errorMessage": "The error message details"}

HTTP status code	Description
400 - Bad Request	One of the input parameters is invalid. See error message in the response body for details.
401 - Unauthorized	The authorization for the request failed. See error message in the response body for details.
403 - Forbidden	The User Access Token in the request header is unknown. This could be the result of a malformed token or a token that has been invalidated by the user removing their consent from the Garmin Connect account page.
412 - Precondition failed	The User Access Token is valid, but the user has not given his permission for the summary-type on the Garmin Connect account page. Other summary-types might still work since the user didn't remove his consent in general
500 - Internal Server Error	Any server error that does not fall in to one of the above categories.

Example

Request:

```
GET https://apis.garmin.com/wellness-api/rest/mct?uploadStartTimeInSeconds=1452384000&uploadEndTimeInSeconds=1452777797000
```

Response:

```
HTTP/1.1 400 Bad Request
Date            Wed, 03 Feb 2016 12:15:17 GMT
Server          Apache
Content-Length 118
Content-Type    application/json;charset=utf-8

{
      "errorMessage": "timestamp '1452777797000' appears to be in milliseconds. Please provide unix timestamps in seconds."
}
```